UNIT-II

Op-Amp, IC 555 & IC 565 Applications

Active Filters

- **Filters** are electronic circuits that allow certain frequency components and/or reject some others. They are passive and are the electric circuits or networks that consist of passive elements like resistors, capacitors, and an inductor.

Types of Active Filters

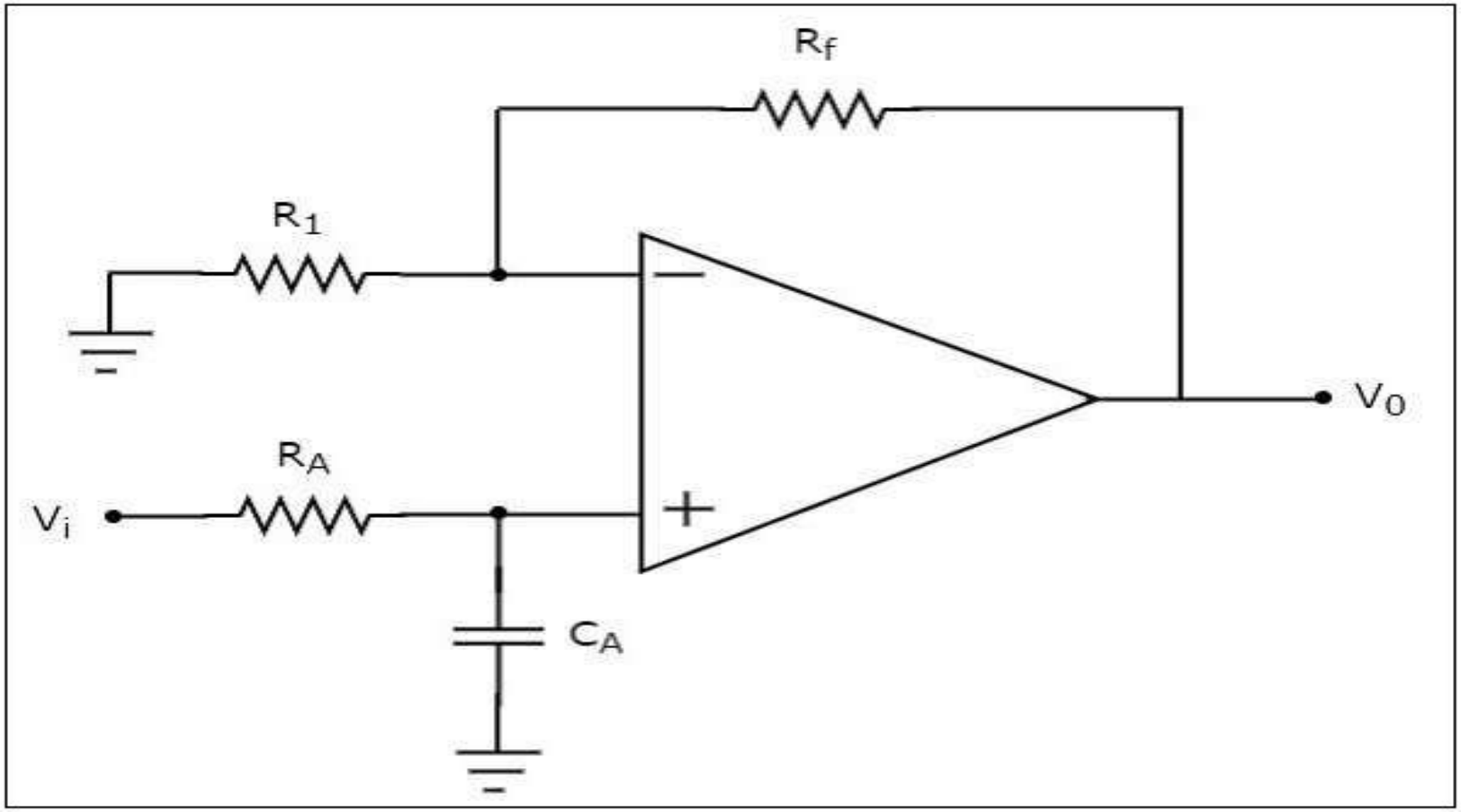
- Active filters are the electronic circuits, which consist of active elements like op-amps along with passive elements like resistors and capacitors.

Active filters are mainly classified into the following **four types** based on the band of frequencies that they allow and/or reject −

- Active Low Pass Filter
- Active High Pass Filter
- Active Band Pass Filter
- Active Band Stop Filter

Active Low Pass Filter

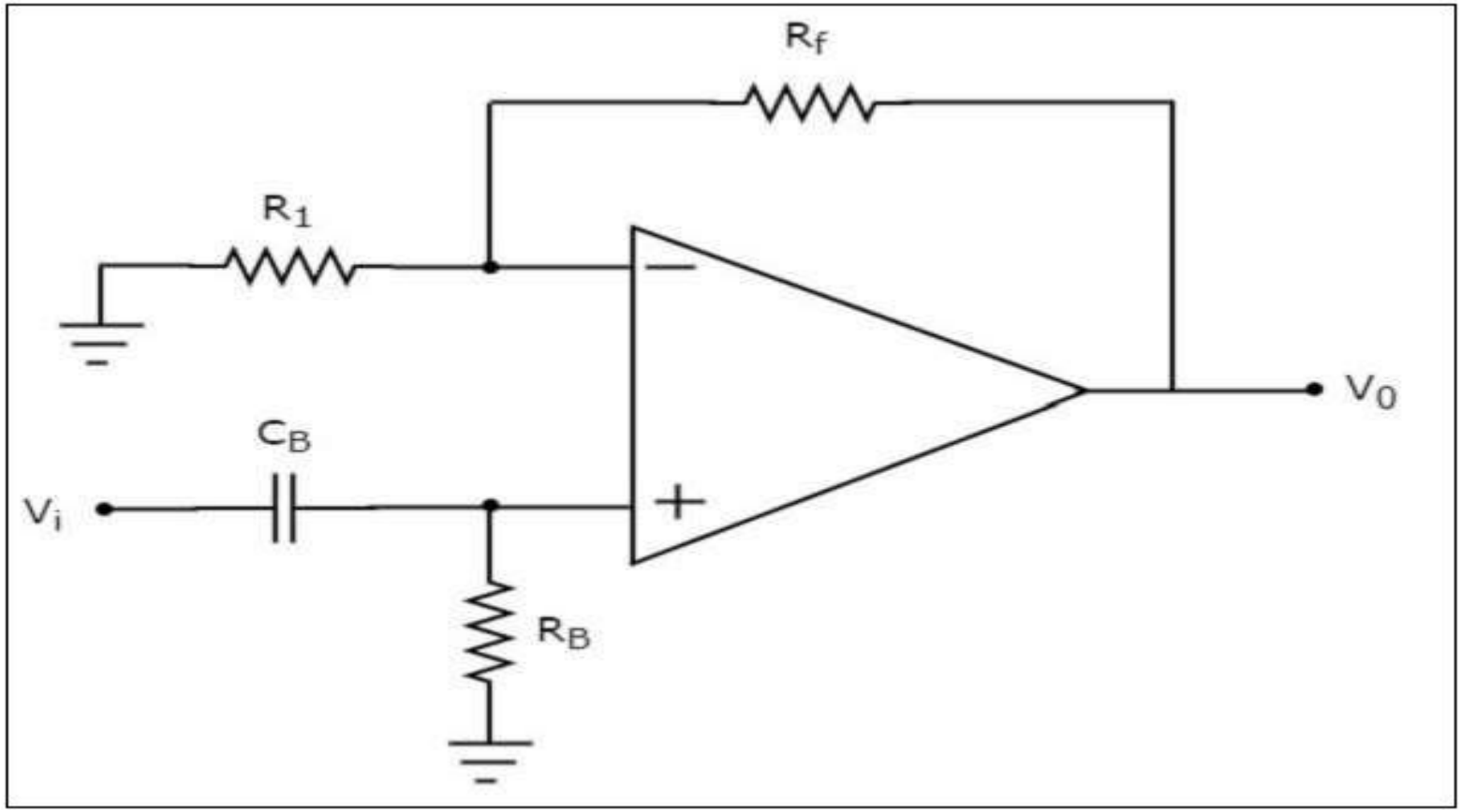
- If an active filter allows (passes) only **low frequency** components and rejects (blocks) all other high frequency components, then it is called an **active low-pass filter**.



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- The electric network, which is connected to the non-inverting terminal of an op-amp, is a **passive low pass filter**. So, the input of a non-inverting terminal of an op-amp is the output of a passive low pass filter.
- Observe that the above circuit resembles a **non-inverting amplifier**. It has the output of a passive low pass filter as an input to the non-inverting terminal of the op-amp. Hence, it produces an output, which is $\left(1+\frac{R_f}{R_1}\right)$ times the input present at the non-inverting terminal.
- The values of Rf and R1 can be suitably chosen to obtain the desired gain at the output. Suppose, if we consider the resistance values of Rf and R1 as zero ohms and infinity ohms, then the above circuit will produce a unity gain low pass filter output.

Active High Pass Filter

- If an active filter allows (passes) only **high frequency** components and rejects (blocks) all other low frequency components, then it is called an **active high pass filter**.



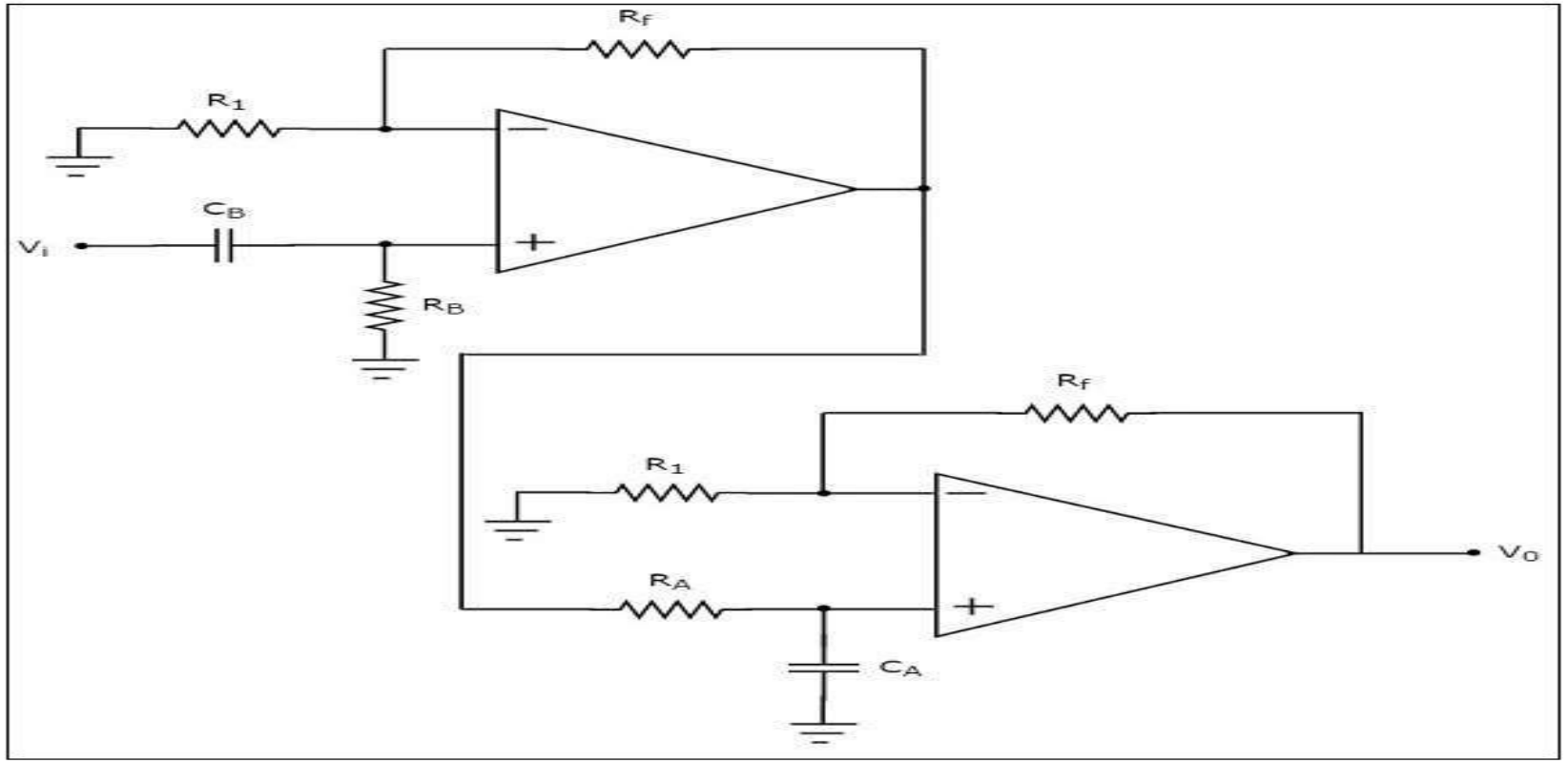
- The electric network, which is connected to the non-inverting terminal of an op-amp, is a passive high pass filter. Therefore, the input of a non-inverting terminal of an op-amp is the output of a passive high-pass filter.

- Now, the above circuit resembles a non-inverting amplifier. It has the output of a passive high-pass filter as its input to the non-inverting terminal of the op-amp. Hence, it produces an output, which is (1+RfR1) times the input present at its non-inverting terminal.

- The values of Rf and R1 can be suitably chosen to obtain the desired gain at the output. Suppose, if we consider the resistance values of Rf and R1 as zero ohms and infinity ohms, then the above circuit will produce a unity gain high pass filter output.

Active Band Pass Filter

- If an active filter allows (passes) only one band of frequencies, then it is called an **active band pass filter**. In general, this frequency band lies between the low-frequency range and the high-frequency range. So, an active band pass filter rejects (blocks) both low and high frequency components.



- There are two parts in the circuit diagram of an active band pass filter: The first part is an active high pass filter, while the second part is an active low pass filter.

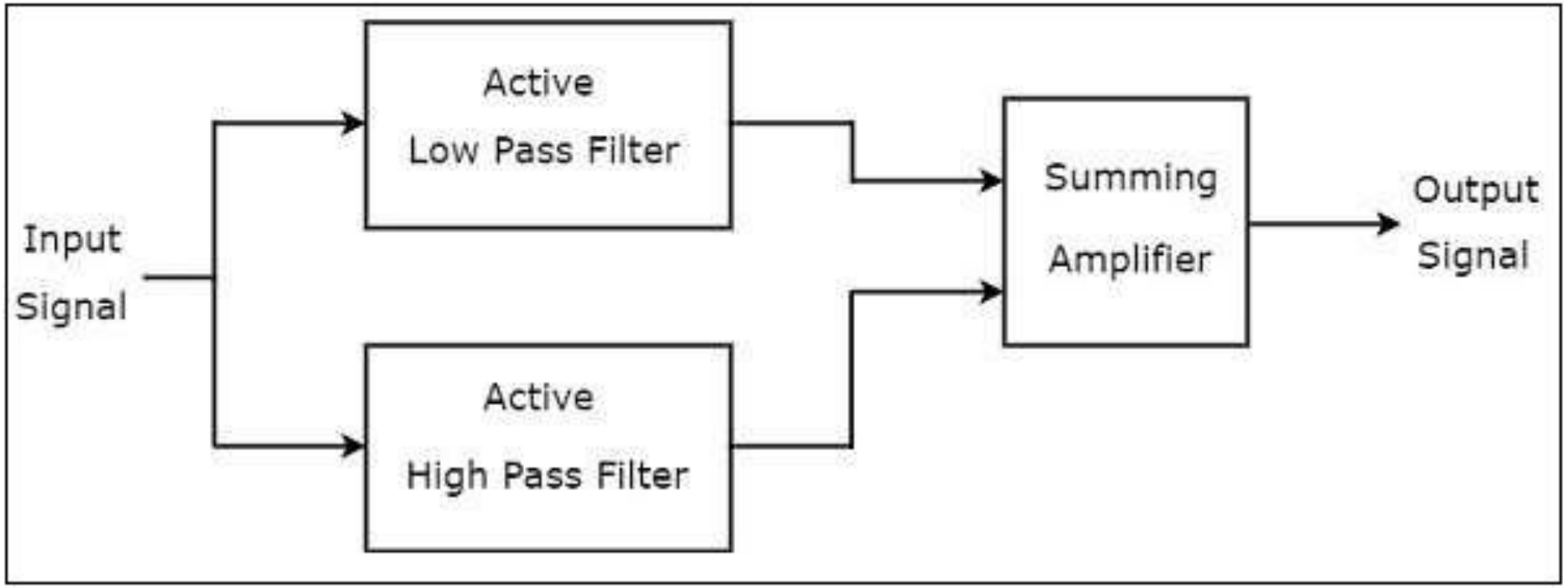
- The output of the active high pass filter is applied as an input to the active low pass filter. That means, both active high pass filter and active low pass filter are cascaded in order to obtain the output in such a way that it contains only a particular band of frequencies.

- The active high pass filter, which is present at the first stage, allows the frequencies that are greater than the lower cut-off frequency of the active band pass filter. So, we have to choose the values of RB and CB suitably to obtain the desired lower cut-off frequency of the active band pass filter.

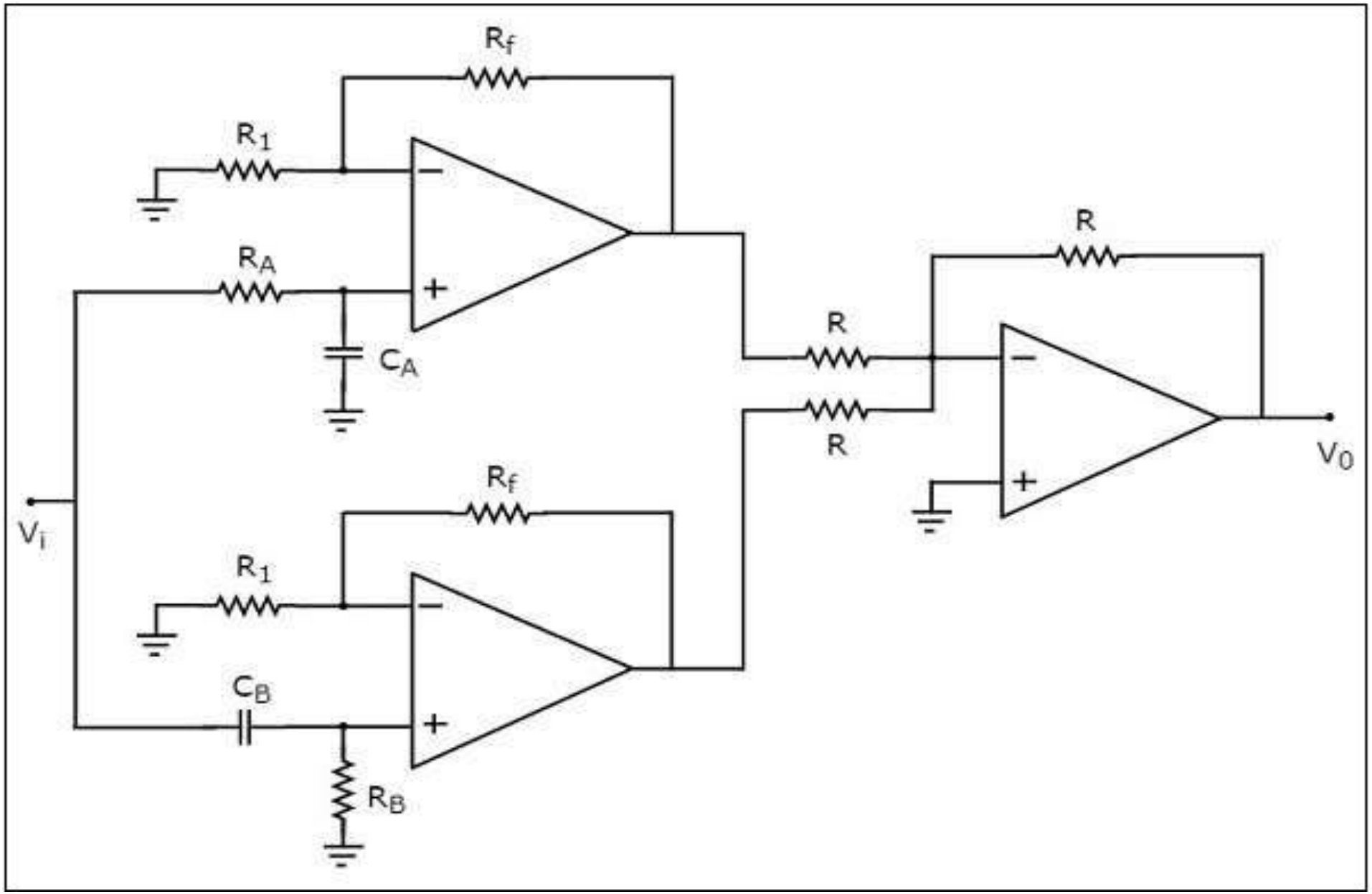
- Similarly, the active low pass filter, which is present at the second stage, allows the frequencies that are smaller than the higher cut-off frequency of the active band pass filter.
- So, the values of RA and CA are suitably chosen to obtain the desired higher cut-off frequency of the active band pass filter.

Active Band Stop Filter

- If an active filter rejects (blocks) a particular band of frequencies, then it is called an **active band stop filter**.
- In general, this frequency band lies between the low frequency range and high frequency range. So, an active band stop filter allows (passes) both low and high frequency components.

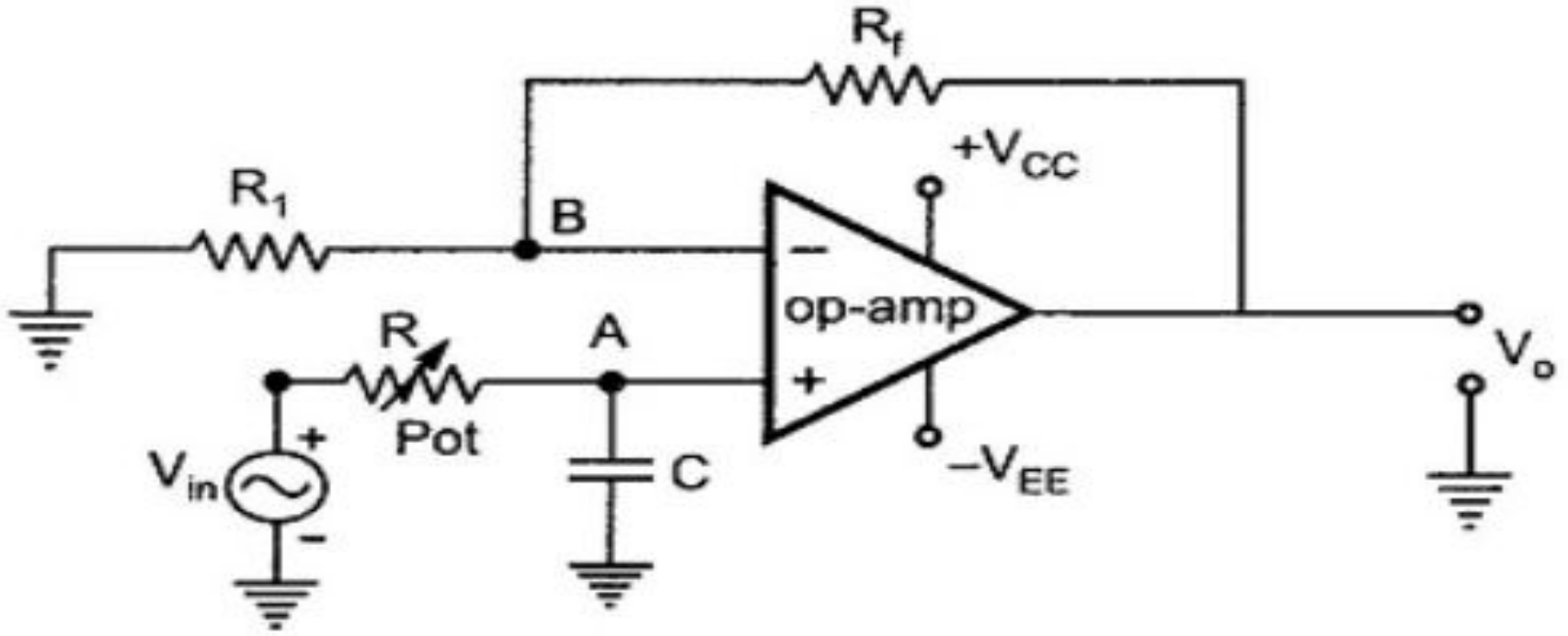


- An active band stop filter consists of two blocks in its first stage: an active low pass filter and an active high pass filter.
- The outputs of these two blocks are applied as inputs to the block that is present in the second stage. So, the **summing amplifier** produces an output, which is the amplified version of the sum of the outputs of the active low pass filter and the active high pass filter.
- Therefore, the output of the above block diagram will be the **output of an active band stop**, when we choose the cut-off frequency of low pass filter to be smaller than the cut-off frequency of a high pass filter.


R_f
R_1
R_A
C_A
V_i
R_f
R_1
C_B
R_B
R
R
R
V_0

First Order Low Pass Butterworth Filter:

- The first order low pass Butterworth filter is realized by an R-C circuit along with an op-amp, used in the noninverting configuration. This is also called one pole low pass Butterworth filter. The resistances R_f and R_1 decide the gain of the filter in the pass band.



Analysis for First Order Low Pass Butterworth Filter Circuit:

- By the potential divider rule, the voltage at the noninverting input terminal A is $-jX_C$ which is the voltage across capacitor C, is given as

$$X_C = \frac{1}{2\pi f C}$$

By the potential divider rule, the voltage at the noninverting input terminal A which is the voltage across capacitor C is given by,

$$V_A = \frac{-jX_C}{R - jX_C} V_{in} \quad \ldots (1)$$

$$V_A = \frac{-j\left(\frac{1}{2\pi f C}\right)}{R - j\left(\frac{1}{2\pi f C}\right)} \cdot V_{in} = \frac{-j}{2\pi f R C - j} \cdot V_{in}$$

$$= \frac{V_{in}}{1 - \frac{2\pi f R C}{j}}$$

but $\quad -j = \frac{1}{j} \quad$ and $\quad -\frac{1}{j} = j$

$$\therefore \quad V_A = \frac{V_{in}}{1 + j 2\pi f R C} \qquad \ldots (2)$$

As the op-amp is in the noninverting configuration,

$$V_o = \left(1 + \frac{R_f}{R_1}\right) V_A \qquad \ldots (3)$$

$$V_o = \left(1 + \frac{R_f}{R_1}\right) \frac{V_{in}}{(1 + j2\pi fRC)}$$

$$\frac{V_o}{V_{in}} = \frac{A_F}{1 + j\left(\frac{f}{f_H}\right)} \qquad \text{... (4)}$$

where

$$A_F = \left(1 + \frac{R_f}{R_1}\right) = \text{gain of filter in pass band} \qquad \text{... (5)}$$

$$f_H = \frac{1}{2\pi RC} = \text{high cut off frequency of filter} \qquad \text{... (6)}$$

The V_o/V_{in} is the transfer function of the filter and can be expressed in the polar form as,

$$\frac{V_o}{V_{in}} = \left|\frac{V_o}{V_{in}}\right| \angle \Phi$$

- Where

$$\left|\frac{V_o}{V_{in}}\right| = \frac{A_F}{\sqrt{1+\left(\frac{f}{f_H}\right)^2}} \qquad \ldots (7)$$

- And

$$\Phi = -\tan^{-1}\left(\frac{f}{f_H}\right) \qquad \ldots (8)$$

1. At very low frequencies, $f < f_H$

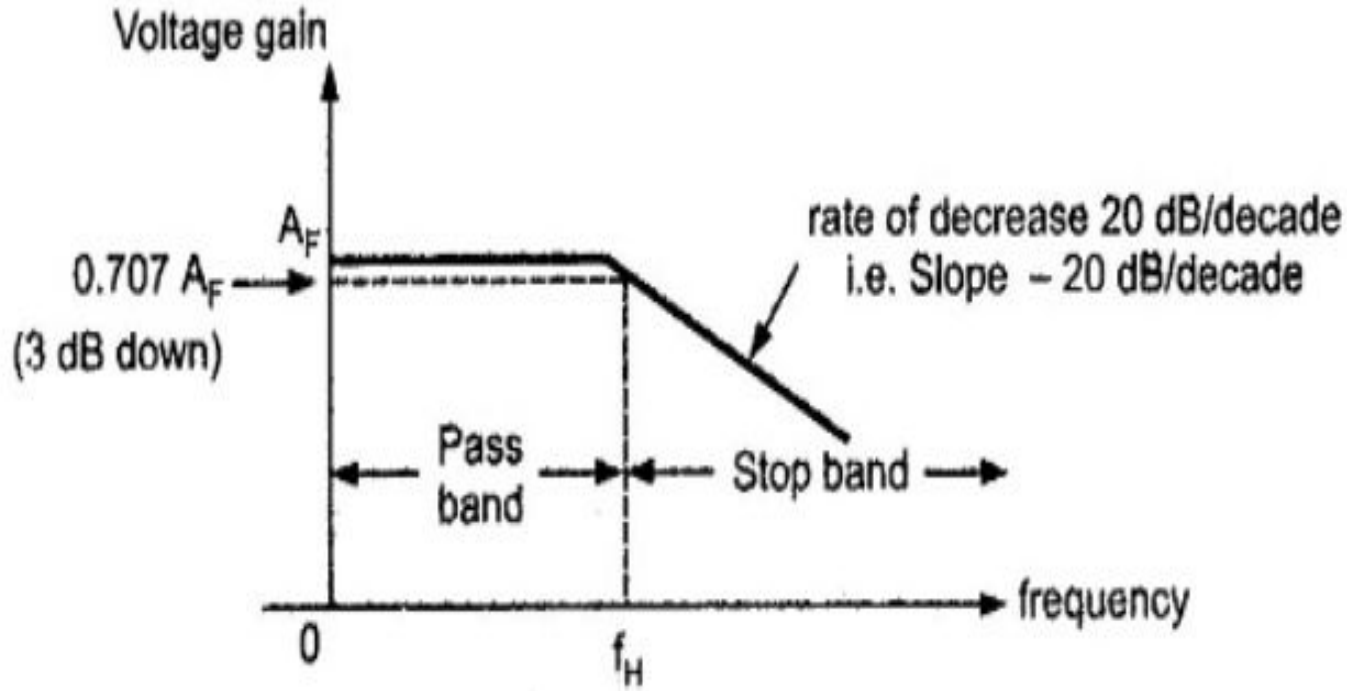
$$\left|\frac{V_o}{V_{in}}\right| \cong A_F \quad \text{i.e. constant}$$

2. $f = f_H$,

$$\left|\frac{V_o}{V_{in}}\right| = \frac{A_F}{\sqrt{2}} = 0.707 \quad A_F$$

3. At $f > f_H$

$$\left|\frac{V_o}{V_{in}}\right| < A_F$$


Voltage gain
A_F
0.707 A_F
(3 dB down)
rate of decrease 20 dB/decade
i.e. Slope – 20 dB/decade
Pass band
Stop band
frequency
0
f_H

- Thus, for the range of frequencies, $0 < f < f_H$, the gain is almost constant, equal to f_H , which is the high cut-off frequency.
- At $f = f_H$, gain reduces to 0.707 A_F i.e., 3 dB down from A_F. And as the frequency increases above f_H, the gain decreases at a rate of 20dB/decade.
- The rate 20 dB/decade means a decrease of 20 dB in gain per 10 times change in frequency.
- The same rate can be expressed as 6 dB/octave, i.e. decrease of 6 dB per two times change in the frequency.
- The frequency f_H is called **cut off frequency**, **break frequency**, — 3dB frequency, or **corner frequency**.

Design Steps:

- The design steps for the first order low pass Butterworth filter are
- 1) Choose the cut-off frequency, fH.
- 2) Choose the capacitance C, usually between 0.001 and 1 µF. Generally, it is selected as 1 µF or less than that. For better performance, mylar or tantalum capacitors are selected.
- 3) Now, for the RC circuit,

$$f_H = \frac{1}{2\pi R C} \qquad \text{refer equation (6)}$$

Hence, as f_H and C are known, calculate the value of R.

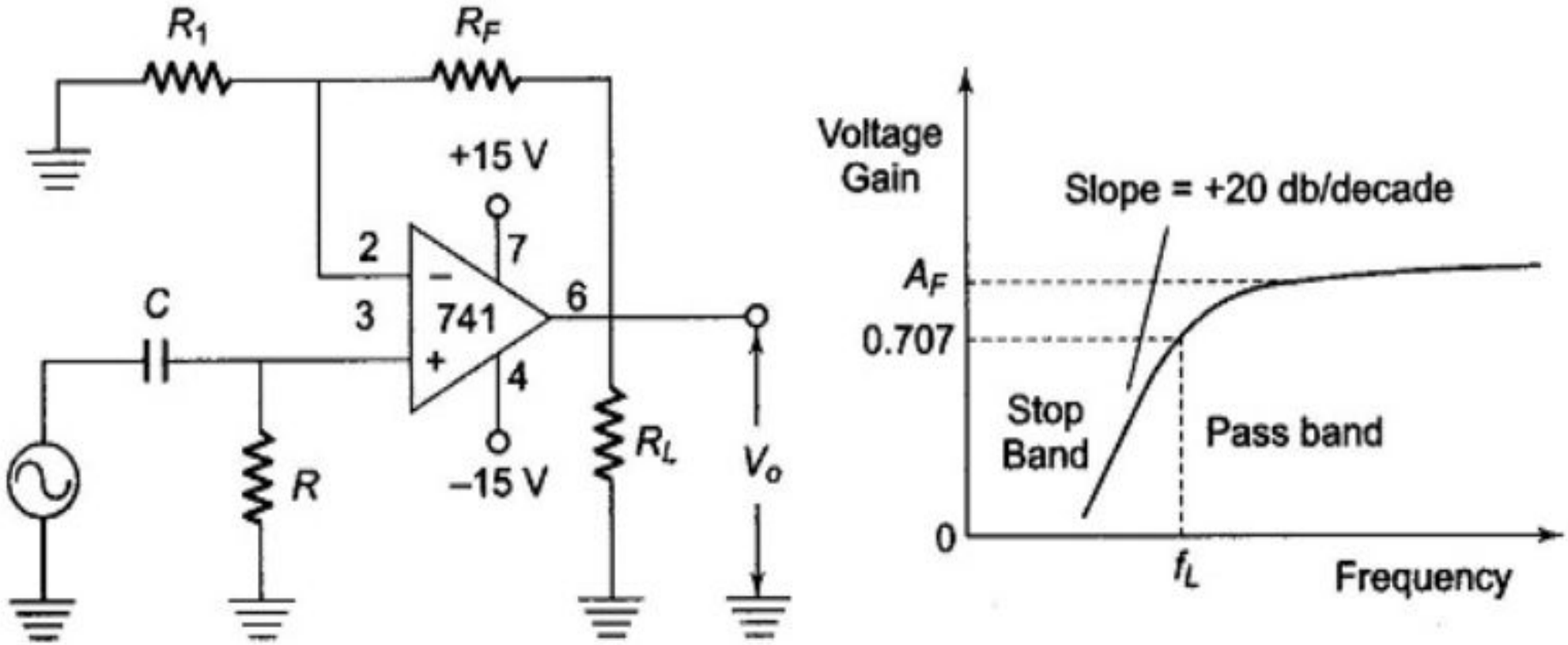
4) The resistances R_f and R_1 can be selected depending on the required gain in the pass band.

$$A_F = 1 + \frac{R_f}{R_1}$$

- Once the filter is designed, sometimes it is necessary to change the value of the cut-off frequency f_H.
- The method used to change the original cut-off frequency f_H to a new cut-off frequency f_{H1} is called **frequency scaling**.
- To achieve such a frequency scaling, the standard value capacitor C is selected first.
- The required cut-off frequency can be achieved by calculating the corresponding value of resistance R. But to achieve frequency scaling, a potentiometer is used.

First Order High Pass Butterworth Filter

- High pass filters are often formed by interchanging frequency determining resistors and capacitors in a low pass filter.
- A first order high pass Butterworth filter with a low cutoff frequency fL, the frequency at which the magnitude of the gain is 0.707 times its passband value.
- All frequencies higher than fL are in the pass band, with the highest frequency determined by the closed loop band width of the op-amp.



(a) (b)

(a) First Order High Pass Butterworth Filter Circuit
(b) First Order High Pass Butterworth Filter Frequency Response

- For the first order high pass filter, the output voltage is

$$V_o = \left(1 + \frac{R_F}{R_1}\right) \times \left(\frac{j2\pi f RC}{1 + j2\pi f RC}\right) \times V_{in}$$

$$\frac{V_o}{V_{in}} = A_F \times \left(\frac{j(f/f_L)}{1 + j(f/f_L)}\right)$$

where

- $A_F = 1 + R_F/R_1$ – pass band gain of the filter
- f = frequency of the input signal
- $f_L = 1/(2 \pi R C)$ – low cutoff frequency

Hence, the magnitude of the voltage gain is

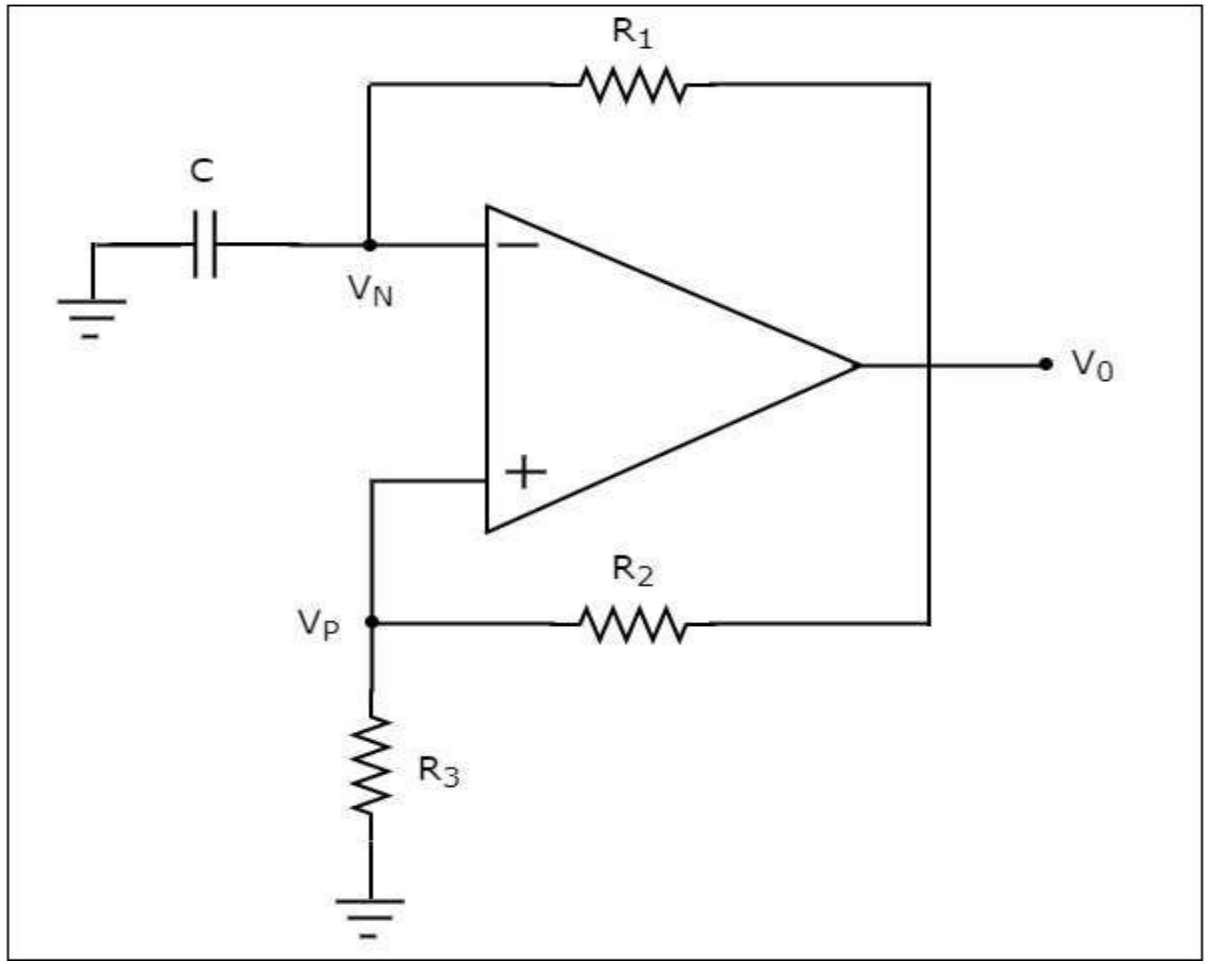
$$\left|\frac{V_o}{V_{in}}\right| = \frac{A_F(f/f_L)}{\sqrt{1 + (f/f_L)^2}}$$

Waveform Generators

- A **waveform generator** is an electronic circuit that generates a standard wave. There are two types of op-amp based waveform generators −
- Square wave generator
- Triangular wave generator

Square Wave Generator

- A **square wave generator** is an electronic circuit that generates a square wave.

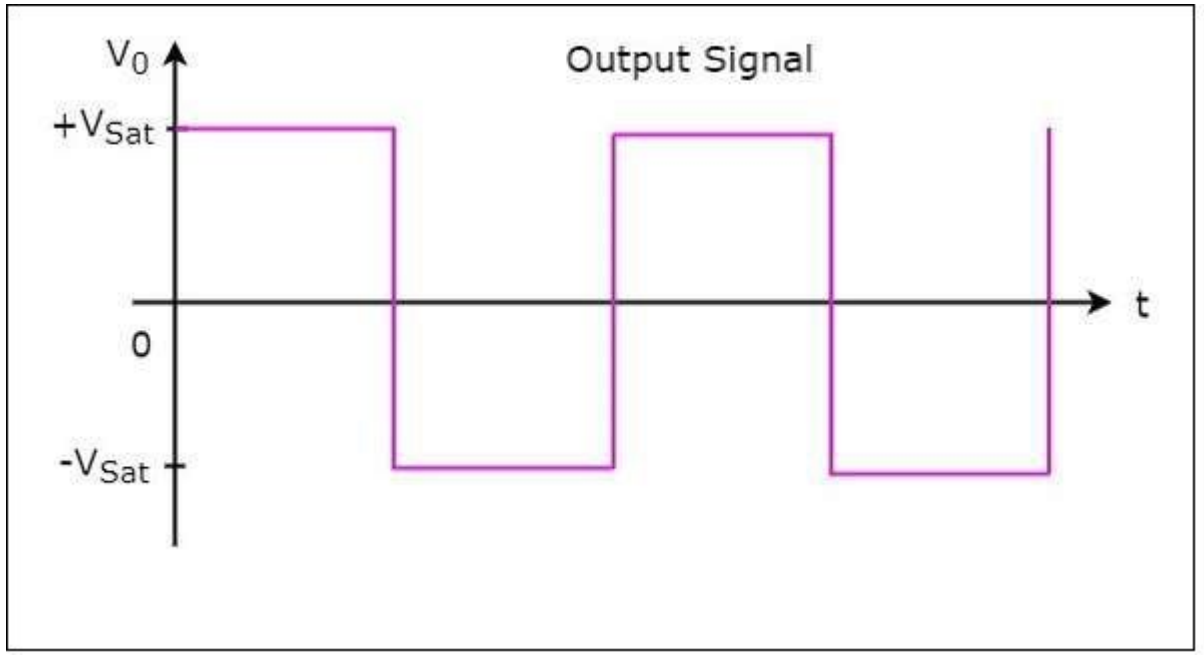


- The resistor R1 is connected between the inverting input terminal of the op-amp and the output of the op-amp. Therefore, resistor R1 is used in the negative feedback.
- Similarly, resistor R2 is connected between the non-inverting input terminal of the op-amp and its output. Therefore, resistor R2 is used in the positive feedback path.
- A capacitor C is connected between the inverting input terminal of the op-amp and ground. So, the voltage across capacitor C will be the input voltage at this inverting terminal of the op-amp.
- Similarly, a resistor R3 is connected between the non-inverting input terminal of the op-amp and ground.
- So, the voltage across resistor R3 will be the input voltage at this non-inverting terminal of the op-amp.

The **operation** of a square wave generator

- Assume there is no charge stored in the capacitor initially. Then, the voltage present at the inverting terminal of the op-amp is zero volts. But there is some offset voltage at the non-inverting terminal of the op-amp. As a result, the value at the output of the above circuit will be +Vsat.
- Now, the capacitor C starts charging through a resistor R1. The value present at the output of the above circuit will change to −Vsat , when the voltage across the capacitor C reaches just greater than the voltage (positive value) across resistor R3.

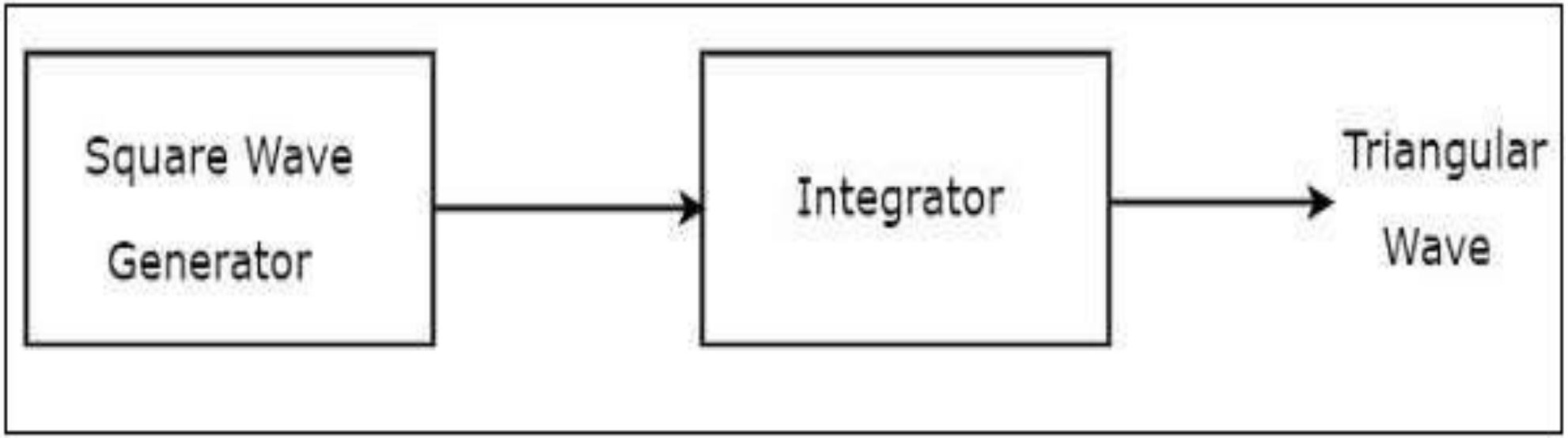
- The capacitor C starts discharging through a resistor R1 when the output of the above circuit is −Vsat . The value present at the output of the above circuit will change to +Vsat when the voltage across capacitor C reaches just less than (more negative) the voltage (negative value) across resistor R3.

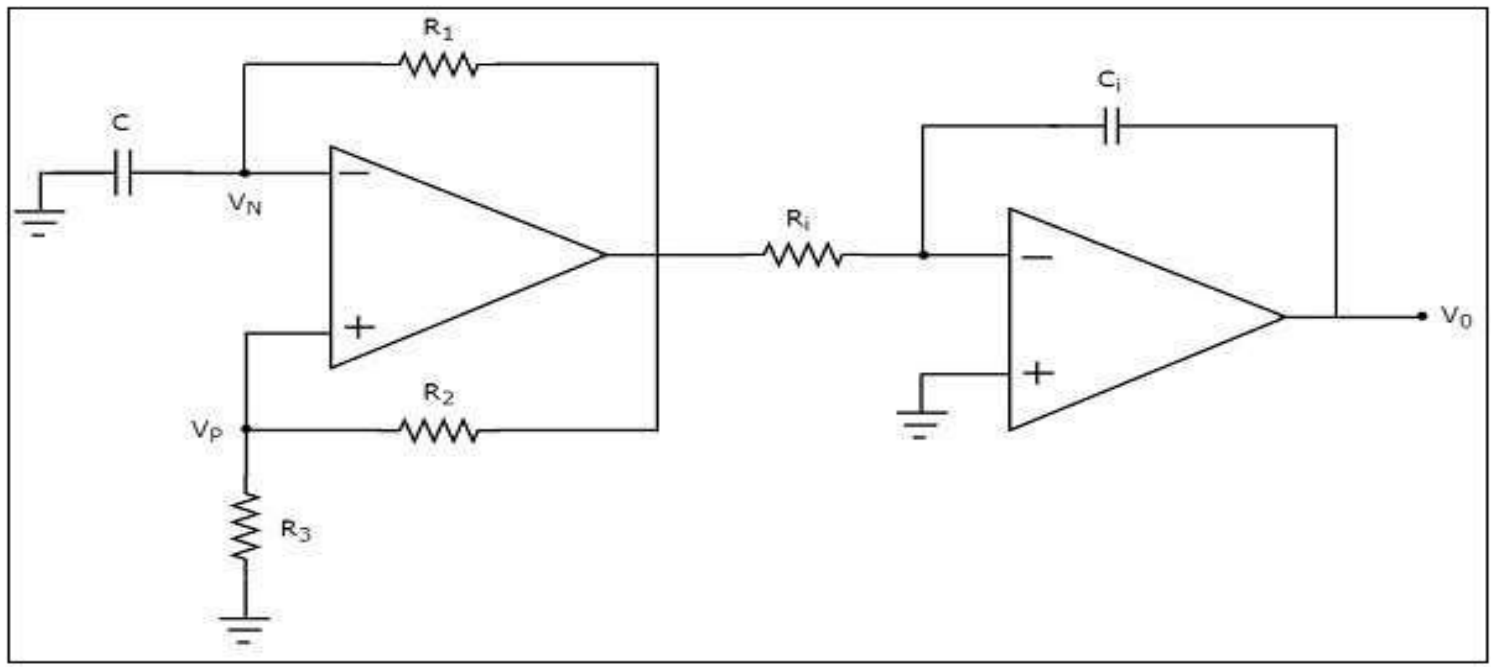


- The output of the square wave generator will have one of the two values: +Vsat and −Vsat. So, the output remains at one value for some duration and then transitions to another value and remains there for some duration.

Triangular Wave Generator

- A triangular wave generator is an electronic circuit that generates a triangular wave.
- The block diagram of a triangular wave generator contains mainly two blocks: a square wave generator and an integrator. These two blocks are **cascaded**. That means, the output of the square wave generator is applied as an input of integrator. The integration of a square wave is nothing but a triangular wave.




R1
C
VN
Ri
Ci
V0
R2
VP
R3

- Basically, a triangular wave is generated by alternatively charging and discharging a capacitor with a constant current.
- This is achieved by connecting the integrator circuit at the output of the square wave generator. Assume that V' is high at $+V_{sat}$.
- This forces a constant current ($+V_{sat}/R_3$) through C (left to right) to drive V_o negative linearly.
- When V' is low at $-V_{sat}$, it forces a constant current ($-V_{sat}/R_3$) through C (right to left) to drive V_o positive, linearly.
- The frequency of the triangular wave is the same as that of **the** square wave.
- Although the amplitude of the square wave is constant ($\pm V_{sat}$), the amplitude of the triangular wave decreases with an increase in its frequency, and vice versa.
- This is because the reactance of a capacitor decreases at high frequencies and increases at low frequencies.
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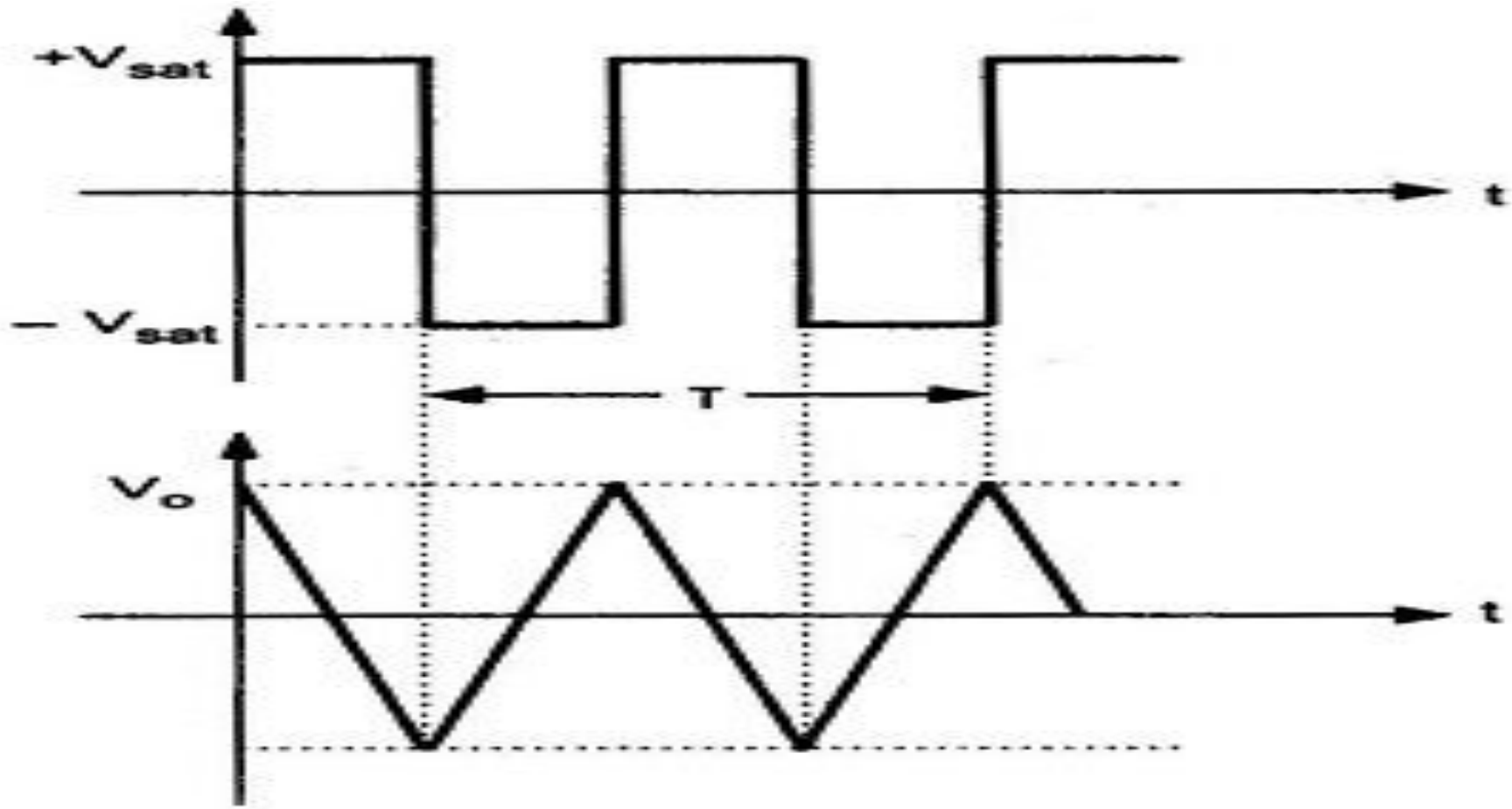
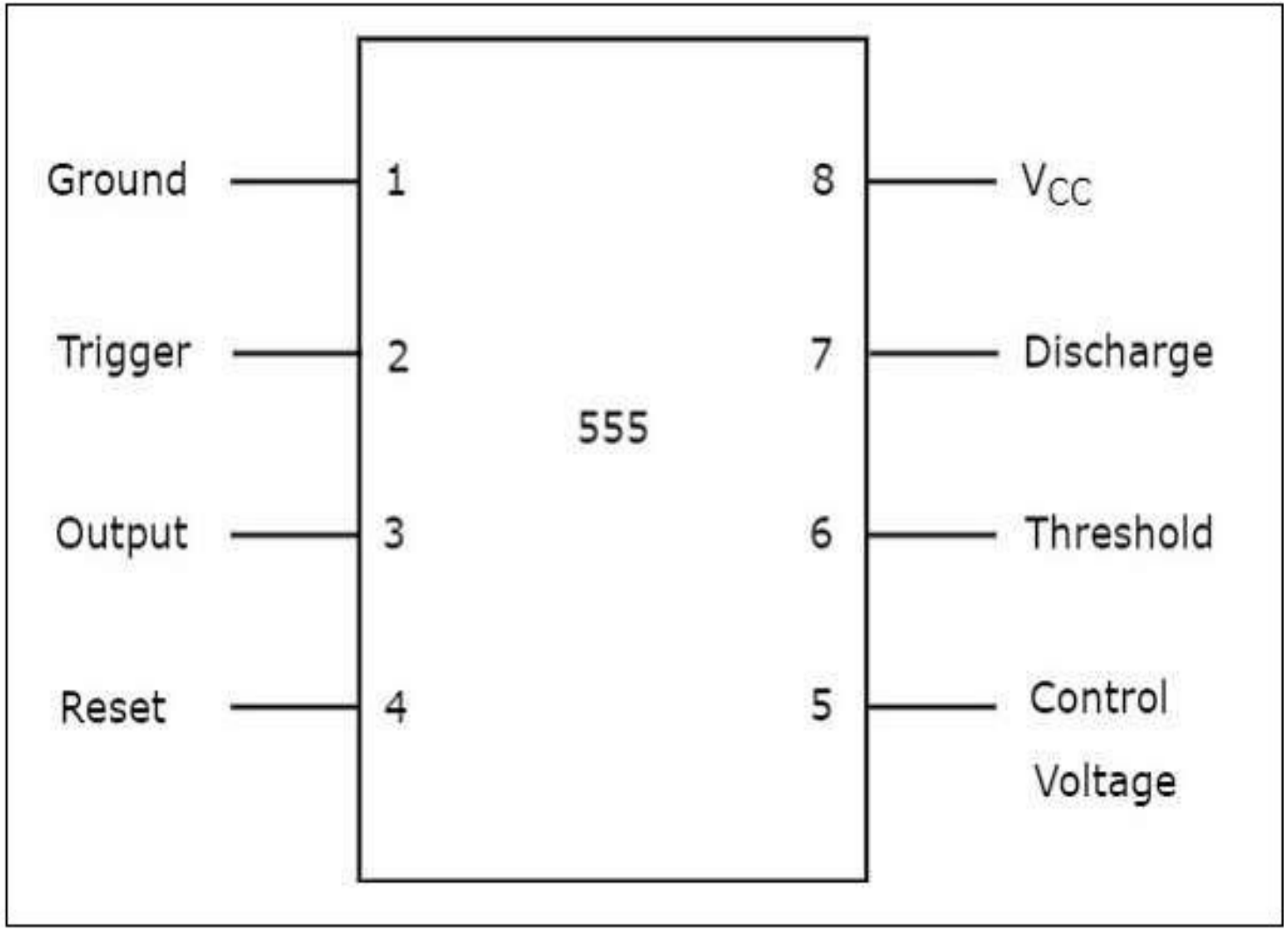


Fig. 2.86 Waveforms of triangular wave generator

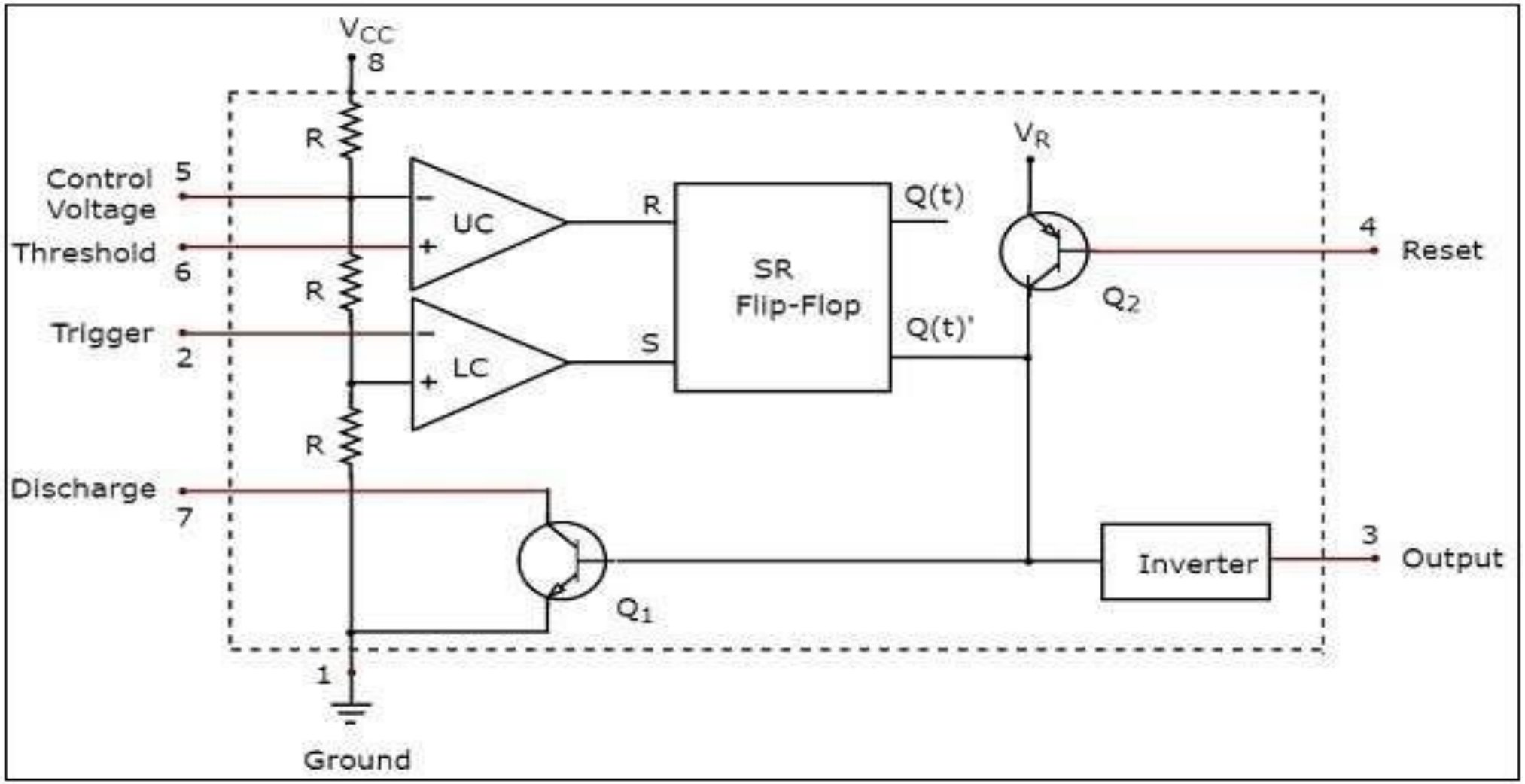
IC 555 Timer

- The 555 Timer IC got its name from the three 5KΩ resistors that are used in its voltage divider network.
- This IC is useful for generating accurate time delays and oscillations.
- This 555 Timer IC can be operated with a DC supply of +5V to +18V.
- It is mainly useful for generating **non-sinusoidal** wave forms like square, ramp, pulse & etc.


Ground
1
Trigger
2
Output
3
Reset
4
555
8
V_{CC}
7
Discharge
6
Threshold
5
Control
Voltage

- Pin 1. – **Ground**, The ground pin connects the 555 timer to the negative (0v) supply rail.
- Pin 2. – **Trigger**, The negative input to comparator No 1. A negative pulse on this pin “sets” the internal Flip-flop when the voltage drops below 1/3Vcc, causing the output to switch from a “LOW” to a “HIGH” state.
- Pin 3. – **Output**, The output pin can drive any TTL circuit and is capable of sourcing or sinking up to 200mA of current at an output voltage equal to approximately Vcc – 1.5V so small speakers, LEDs, or motors can be connected directly to the output.
- Pin 4. – **Reset**. This pin is used to “reset” the internal Flip-flop controlling the state of the output, pin 3. This is an active-low input and is generally connected to a logic “1” level when not used to prevent any unwanted resetting of the output.
- Pin 5. – **Control Voltage**, This pin controls the timing of the 555 by overriding the 2/3Vcc level of the voltage divider network. By applying a voltage to this pin the width of the output signal can be varied independently of the RC timing network. When not used, it is connected to ground via a 10nF capacitor to eliminate any noise.
- Pin 6. – **Threshold**, The positive input to comparator No 2. This pin is used to reset the Flip-flop when the voltage applied to it exceeds 2/3Vcc, causing the output to switch from “HIGH” to “LOW” state. This pin connects directly to the RC timing circuit.
- Pin 7. – **Discharge**, The discharge pin is connected directly to the Collector of an internal NPN transistor which is used to “discharge” the timing capacitor to ground when the output at pin 3 switches “LOW”.
- Pin 8. – **Supply +Vcc**. This is the power supply pin, and for general-purpose TTL 555 timers is between 4.5V and 15V.

Functional Diagram



- The functional diagram of the 555 Timer contains a voltage divider network, two comparators, one SR flip-flop, two transistors, and an inverter.

Voltage Divider Network

- The voltage divider network consists of three 5KΩ resistors that are connected in series between the supply voltage Vcc and ground.
- This network provides a voltage of Vcc/3 between a point and ground if there exists only one 5KΩ resistor. Similarly, it provides a voltage of 2Vcc/3 between a point and ground if there exist only two 5KΩ resistors.

Comparator

- The functional diagram of a 555 Timer IC consists of two comparators: an Upper Comparator (UC) and a Lower Comparator (LC).
- Recall that a comparator compares the two inputs that are applied to it and produces an output.
- If the voltage present at the non-inverting terminal of an op-amp is greater than the voltage present at its inverting terminal, then the output of the comparator will be +Vsat. This can be considered as Logic High (‘1’) in digital representation.
- If the voltage present at the non-inverting terminal of the op-amp is less than or equal to the voltage at its inverting terminal, then the output of the comparator will be −Vsat. This can be considered as Logic Low ('0') in digital representation.

SR Flip-Flop

- An SR flip-flop operates with either positive clock transitions or negative clock transitions. It has two inputs: S and R, and two outputs: Q(t) and Q(t). The outputs, Q(t) & Q(t), are complementary to each other.
- S R Q(t+1)
- 0 0 Q(t)
- 0 1 0
- 1 0 1
- 1 1 -
- Here, Q(t) & Q(t+1) are present state & next state respectively. So, the SR flip-flop can be used for one of these three functions, such as Hold, Reset & Set based on the input conditions, when positive (negative) transition of the clock signal is applied.

- The outputs of Lower Comparator (LC) and Upper Comparator (UC) are applied as inputs of the SR flip-flop, as shown in the functional diagram of 555 Timer IC.

Transistors and Inverter

- The functional diagram of a 555 Timer IC consists of one npn transistor Q1 and one pnp transistor Q2. The npn transistor Q1 will be turned ON if its base to emitter voltage is positive and greater than the cut-in voltage. Otherwise, it will be turned-OFF.

- The pnp transistor Q2 is used as a buffer to isolate the reset input from the SR flip-flop and npn transistor Q1.

- The inverter used in the functional diagram of a 555 Timer IC not only performs the inverting action but also amplifies the power level.

Monostable Multivibrator Using IC 555

- The IC 555 timer can be operated as a Monostable Multivibrator Using IC 555 by connecting an external resistor and a capacitor

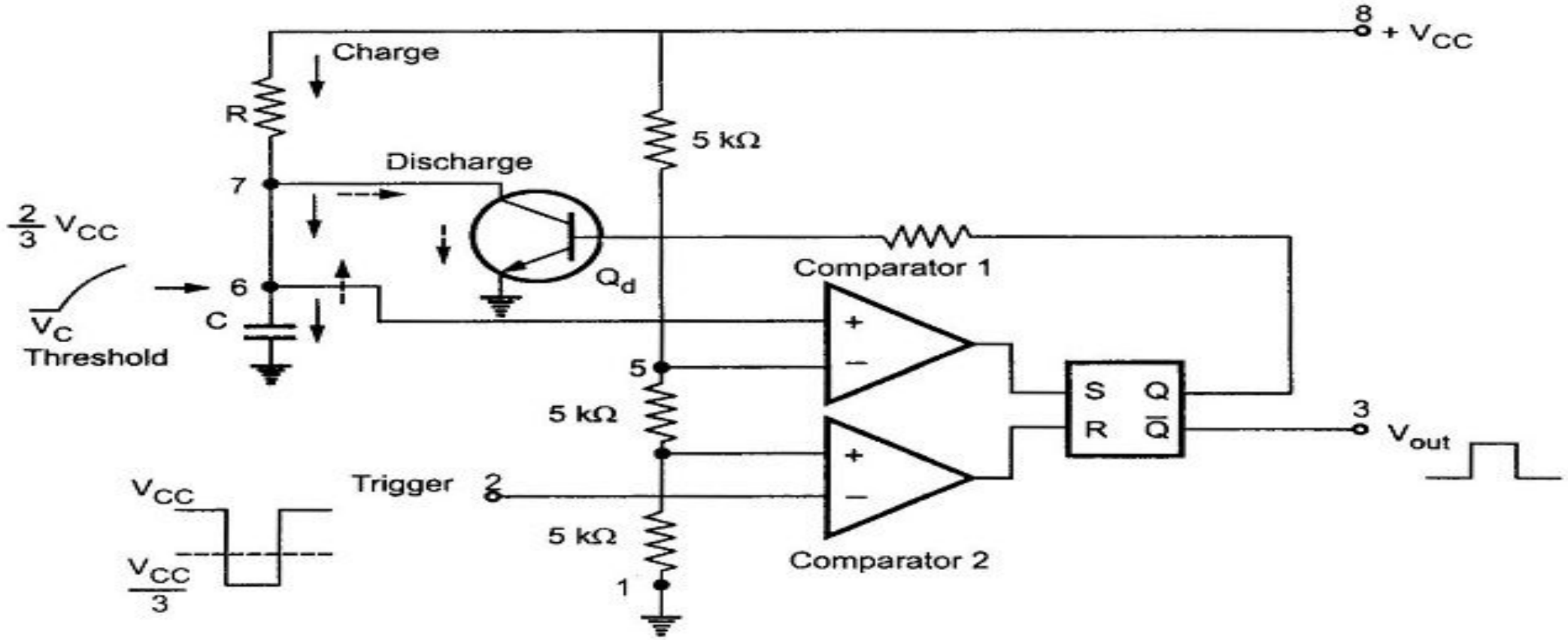


Fig. 2.102 Monostable operation of 555

- The circuit has **only one stable state**. When trigger is applied, it produces a pulse at the output and returns back to its stable state.
- The duration of the pulse depends on the values of R and C. As it has only one stable state, it is called one shot **multivibrator**.

Working of 555 Timer as Monostable Multivibrator:

- The flip-flop is initially set i.e. Q is high. This drives the transistor Q_d in saturation. The capacitor discharges completely and voltage across it is nearly zero. The output at pin 3 is low.
- When a trigger input, a low going pulse is applied, then circuit state remains unchanged till trigger voltage is greater than 1/3 V_{CC}.
- When it becomes less than 1/3 V_{CC}, then comparator 2 output goes high. This resets the flip-flop so Q goes low and $\bar{Q}$ goes high. Low Q makes the transistor Q_d off. Hence capacitor starts charging through resistance R

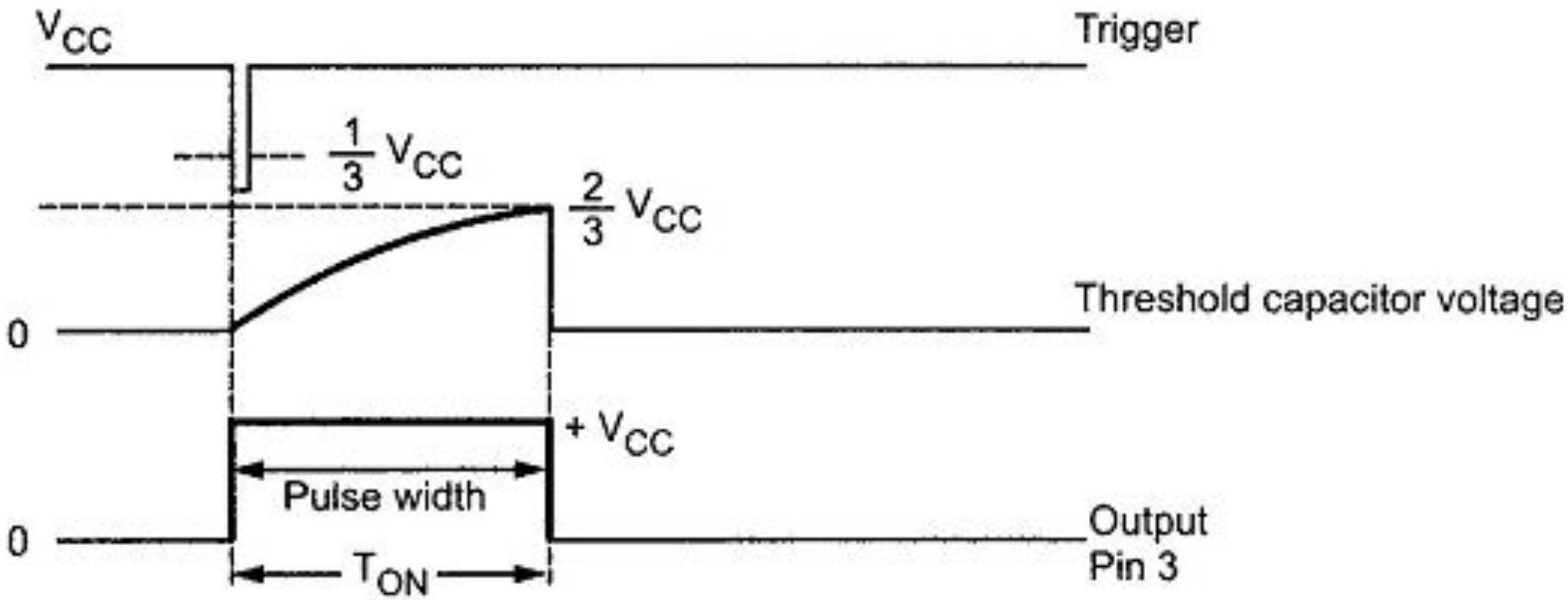


Fig. 2.103 Waveforms of monostable operation

- The voltage across capacitor increases exponentially. This voltage is nothing but the threshold voltage at pin 6. When this voltage becomes more than 2/3 V_{CC}, then comparator 1 output goes high.
- This sets the flip-flop i.e. Q becomes high and $\bar{Q}$ low. This high Q drives the transistor Q_d in saturation. Thus capacitor C quickly discharges through $Q_{d.}$
- So it can be noted that V_{out} at pin 3 is low at start, when trigger is less than 1/3 V_{CC} it becomes high and when threshold is greater than 2/3 V_{CC} again becomes low, till next trigger pulse occurs.
- So a rectangular wave is produced at the output. The pulse width of this rectangular pulse is controlled by the charging time of capacitor. This depends on the time constant RC. Thus RC controls the pulse width.

- **Derivation of Pulse Width**

The voltage across capacitor increases exponentially and is given by

$$V_C = V\,(1 - e^{-t/CR})$$

If $V_C = 2/3\ V_{CC}$

then $\frac{2}{3} V_{CC} = V_{CC}\,(1 - e^{-t/CR})$

$$\frac{2}{3} - 1 = -e^{t/CR}$$

$$\frac{1}{3} = e^{-t/CR}$$

$$\therefore \quad -\frac{t}{CR} = -1.0986$$

$$\therefore \quad t = +1.0986\ CR$$

$$\therefore \quad t \approx 1.1\ CR$$

- **Schematic Diagram:**
- The external components R and C are shown. To avoid accidental reset, pin 4 is connected to pin 8 which is supply $+V_{CC}$. To have the noise filtering of control voltage, the pin 5 is grounded through a small capacitor of 0.01 µF.

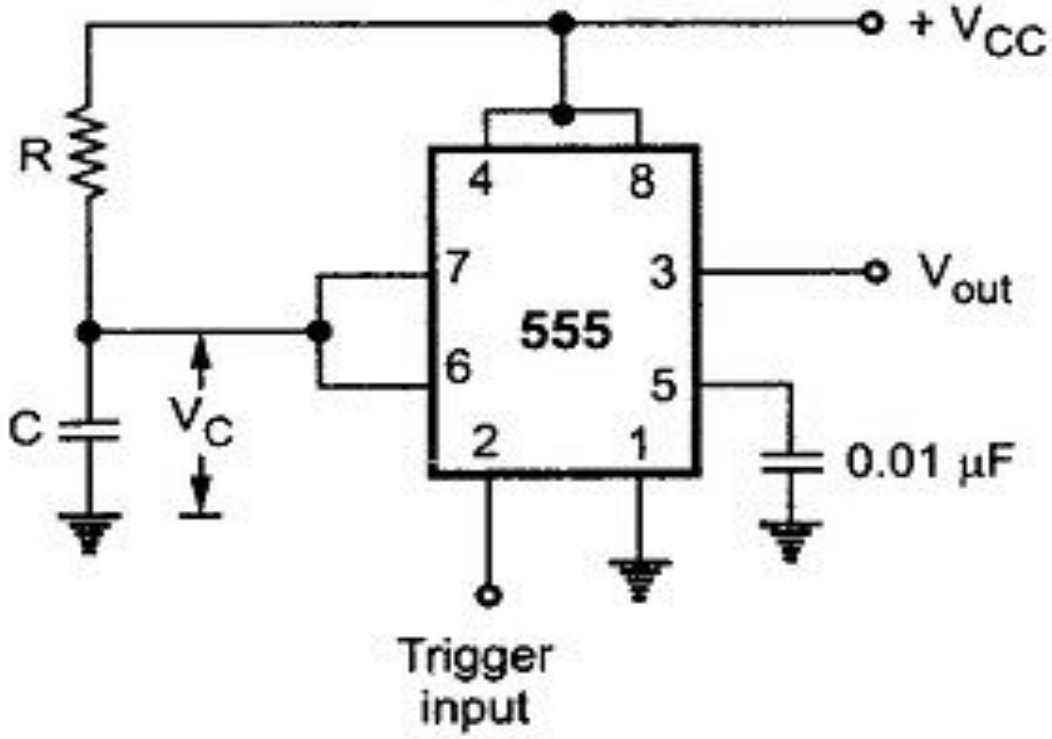


Fig. 2.104 555 timer as monostable multivibrator

- **Monostable Multivibrator Using IC 555 Applications:**

The various applications of Monostable Multivibrator Using IC 555 are,

- Frequency divider
- Pulse width modulation
- Linear ramp generator
- Pulse position modulation
- Missing pulse detector
- Timer in relay

Astable Multivibrator Using IC 555

- The threshold input is connected to the trigger input. Two external resistances R_A, R_B and a capacitor C is used in the circuit.

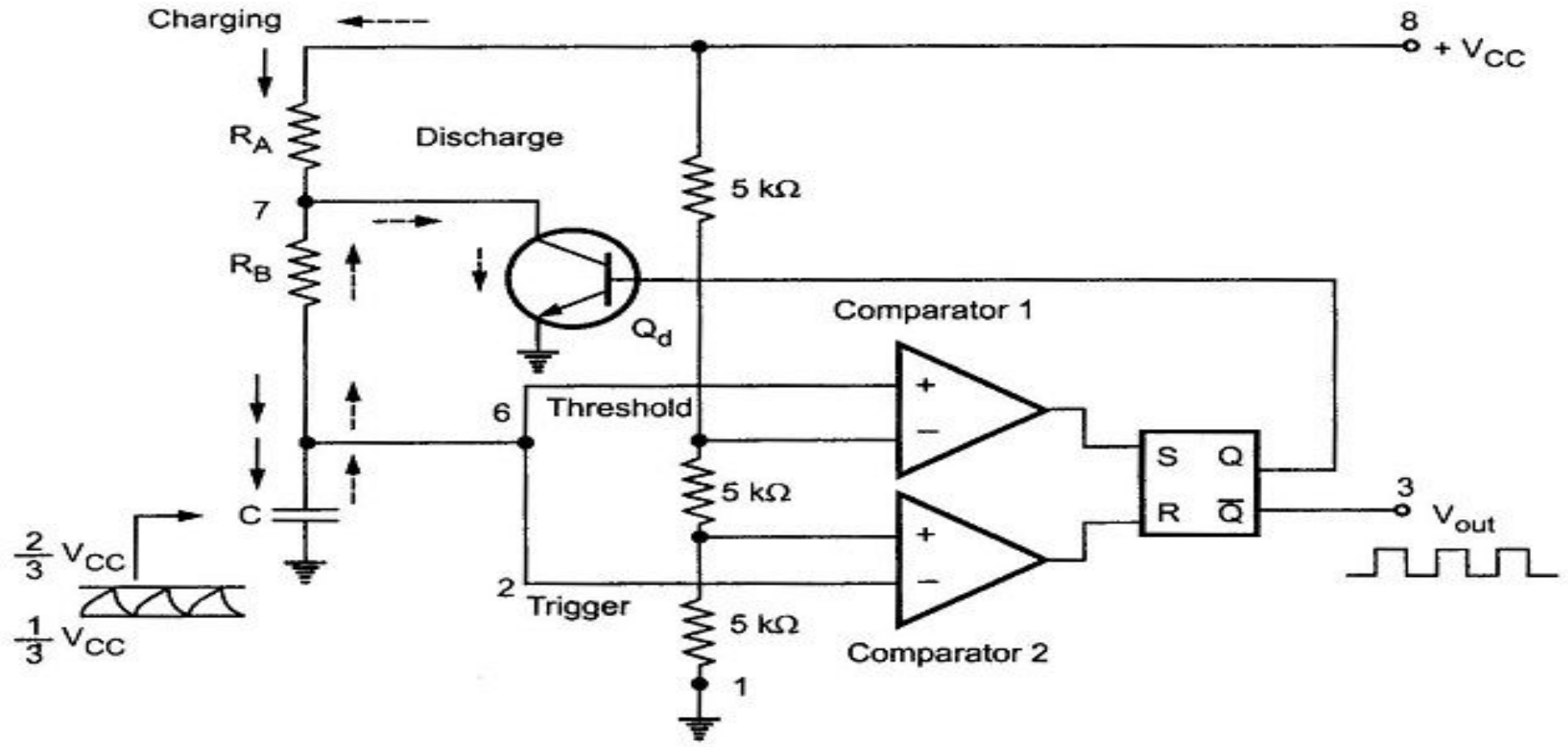


Fig. 2.105 Astable operation of 555

Working of Astable Multivibrator using IC 555:

- When the flip-flop is set, Q is high which drives the transistor Q_d in saturation and the capacitor gets discharged.
- Now the capacitor voltage is nothing but the trigger voltage. So while discharging, when it becomes less than 1/3 V_{CC}, comparator 2 output goes high. This resets the flip-flop hence Q goes low and $\bar{Q}$ goes high.
- The low Q makes the transistor off.
- Thus capacitor starts charging through the resistances R_A, R_B and V_{CC}. As total resistance in the charging path is $(R_A + R_B)$, the charging time constant is $(R_A + R_B)$ C.

- Now the capacitor voltage is also a threshold voltage. While charging, capacitor voltage increases i.e. the threshold voltage increases.
- When it exceeds 2/3 V_{CC}, then the comparator 1 output goes high which sets the flip-flop. The flip-flop output Q becomes high and output at pin 3 i.e. $\bar{Q}$ becomes low.
- High Q drives transistor Q_d in saturation and capacitor starts discharging through resistance R_B and transistor Q_d.
- Thus the discharging time constant is R_B C. When capacitor voltage becomes less than 1/3 V_{CC}, comparator 2 output goes high, resetting the flip-flop. This cycle repeats.
- Thus when capacitor is charging, output is high while when it is discharging the output is low. The output is a rectangular wave. The capacitor voltage is exponentially rising and falling.

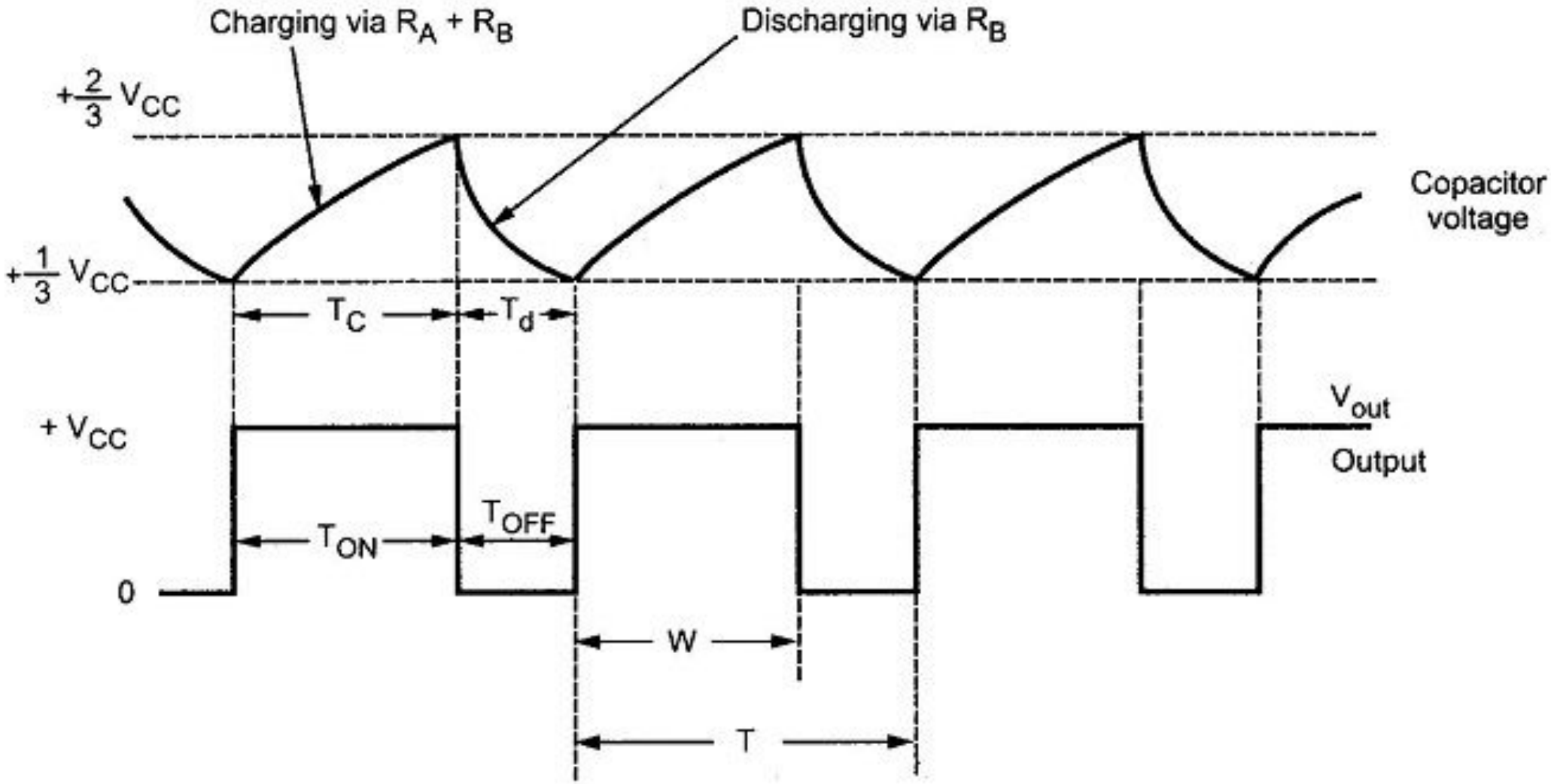


Fig. 2.106 Waveforms of astable operation

- **Duty Cycle of Astable Multivibrator**
- Generally the charging time constant is greater than the discharging time constant. Hence at the output, the waveform is not symmetric.
- The high output remains for longer period than low output. The ratio of high output period and low output period is given by a mathematical parameter called **duty cycle**.
- **It is defined as the ratio of ON time i.e. high output to the total time of one cycle.**

$$W = \text{time for output is high} = T_{ON}$$

$$T = \text{time of one cycle}$$

$$\therefore \quad D = \text{duty cycle} = \frac{W}{T}$$

$$\therefore \quad \%\ D = \frac{W}{T} \times 100\ \%$$

The charging time for the capacitor is given by,

$$T_c = \text{Charging time} = 0.693\ (R_A + R_B)\ C$$

While the discharge time is given by,

$$T_d = \text{Discharging time} = 0.693\ R_B\ C$$

Hence the time for one cycle is

$$T = T_c + T_d = 0.693\ (R_A + R_B)\ C + 0.693\ R_B\ C$$

$$= 0.693\ (R_A + 2\ R_B)\ C$$

$$W = T_c = 0.693\,(R_A + R_B)\,C$$

$$\therefore \quad \%D = \frac{W}{T} \times 100 = \frac{0.693\,(R_A + R_B)\,C}{0.693\,(R_A + 2\,R_B)\,C} \times 100$$

$$\therefore \quad \%\,D = \frac{(R_A + R_B)}{(R_A + 2\,R_B)} \times 100$$

While the frequency of oscillations is given by,

$$f = \frac{1}{T} = \frac{1}{0.693\,(R_A + 2\,R_B)C}$$

$$\therefore \quad f = \frac{1.44}{(R_A + 2\,R_B)C}\ \text{Hz}$$

If R_A is much smaller than R_B, duty cycle approaches to 50% and output waveform approaches to square wave.

- **Schematic Diagram of 555 Timer Astable Multivibrator**
- It shows only the external components R_A, R_B and C. The pin 4 is tied to pin 8 and pin 5 is grounded through a small capacitor.

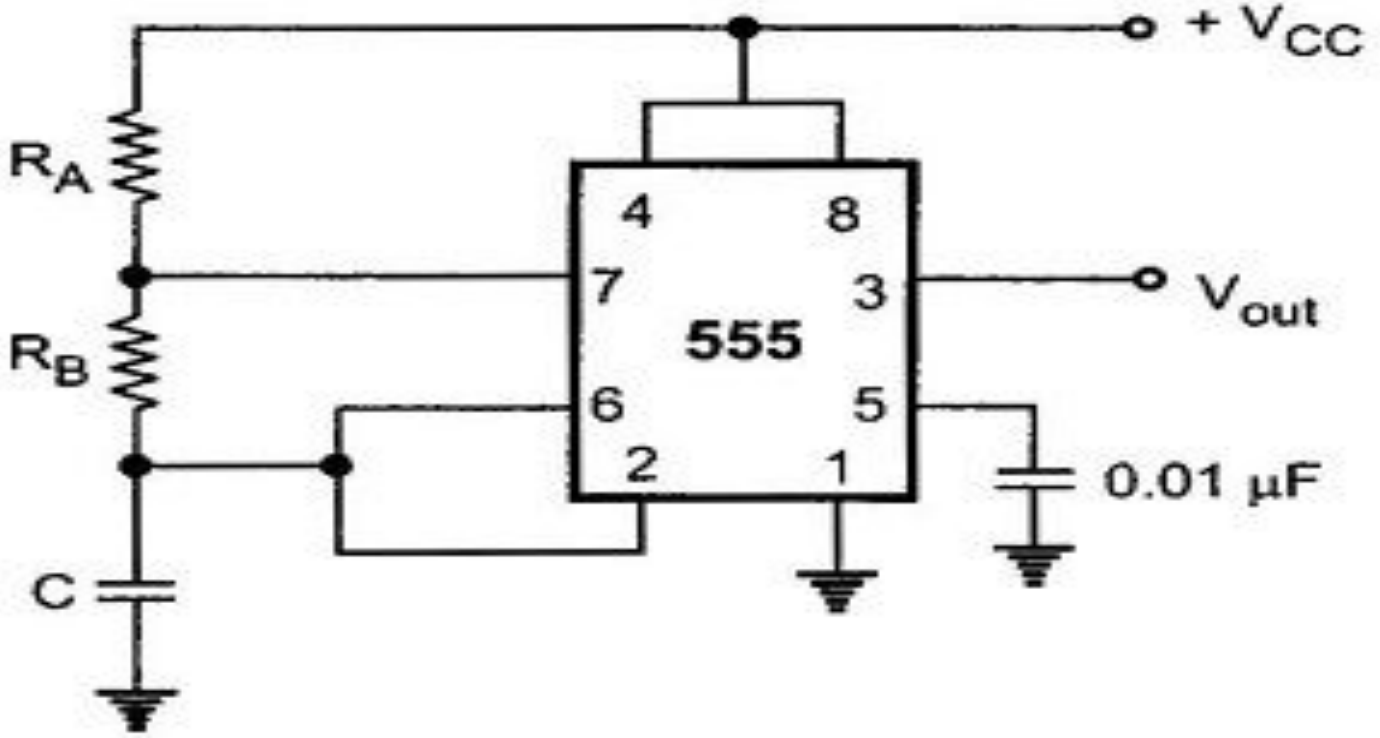


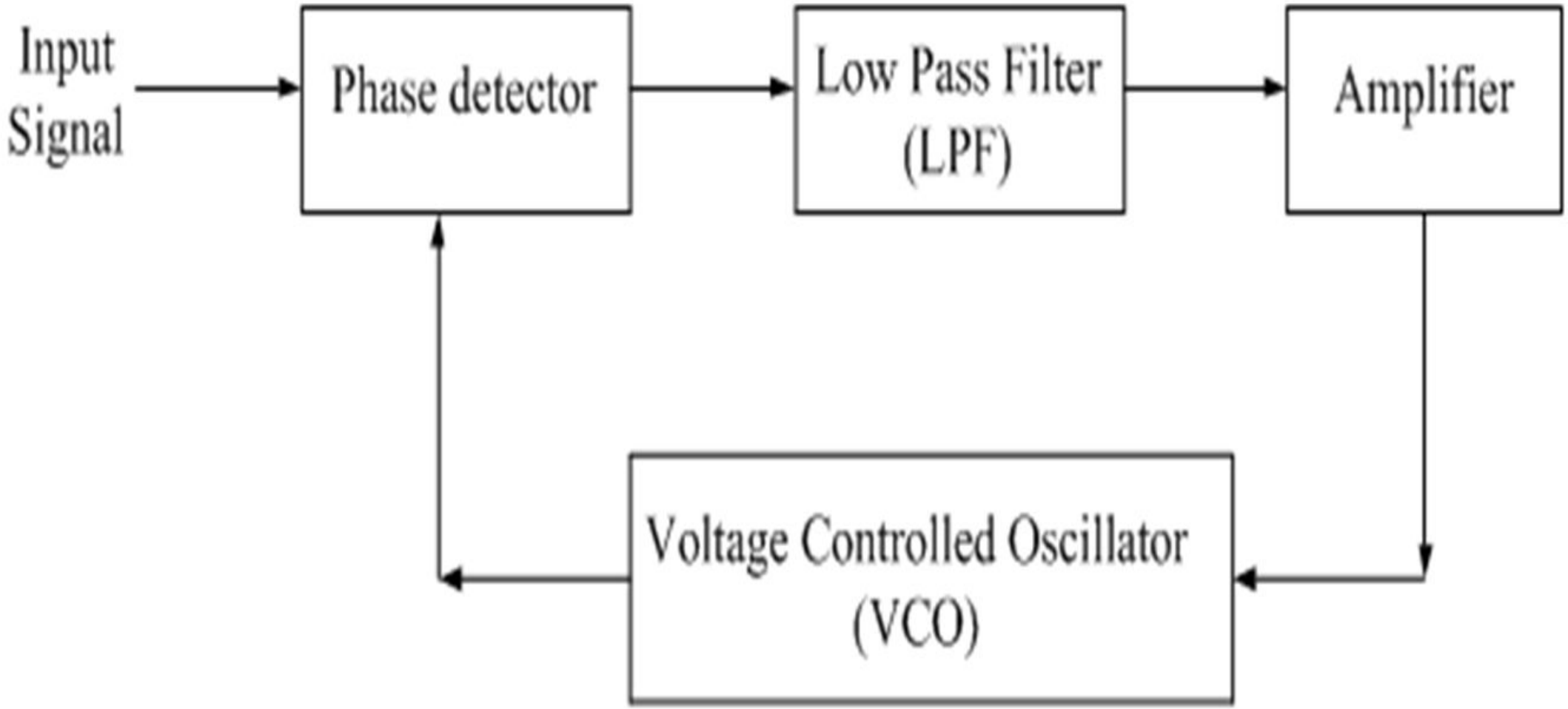
Fig. 2.108 555 timer as astable multivibrator

Application of Astable Multivibrator using IC 555:

- The various Application of Astable Multivibrator using IC 555 are,
- Square wave generation
- FSK generator
- Voltage controlled oscillator (VCO)

Phase Locked Loop **(PLL)**

- A Phase Locked Loop **(PLL)** is one of the vital blocks in linear systems. It is useful in communication systems such as radars, satellites, FMs, etc.
- A Phase Locked Loop (PLL) mainly consists of the following **three blocks** −
- Phase Detector
- Active Low Pass Filter
- Voltage Controlled Oscillator (VCO)


Input Signal
Phase detector
Low Pass Filter (LPF)
Amplifier
Voltage Controlled Oscillator (VCO)

- The output of a phase detector is applied as an input of active low-pass filter. Similarly, the output of the active low-pass filter is applied as an input of VCO.

The working of a PLL is as follows –

- The phase detector block has two inputs, external signal and oscillator signal. Phase detector detects the phase between two signals and develops an output voltage proportional to phase difference.
- This block also called as phase detector or phase comparator. Output of phase detector is passed through a LPF.
- It removes high frequency signal and passes only low frequencies. This filter is also called as loop filter because the PLL system is a closed loop system.

- The amplifier amplifies the output of the filter. This amplified output is applied as input for the VCO. This input voltage adjusts the VCO frequency so that it equals the signal frequency; i.e., the VCO converts an input voltage to a frequency, acting as a voltage-to-frequency converter.
- When the signal frequency and VCO frequency is same, the loop gets locked. The loop gets locked by detecting the phase difference between two inputs, so called Phase Locked Loop (PLL).
- Without application of any external signal, VCO has a frequency called as free running frequency or centre frequency. In this initial condition loop is not locked, i.e., in an open condition.
- When external signal is applied its frequency is either less or greater than VCO frequency so there is a phase difference between them. Phase detector detects the phase difference between two inputs and generates an error voltage. This is passed through LPF.

- After amplification, it is given as a controlled voltage. This adjusts the frequency of the VCO such that the input frequency is equal to the VCO frequency and forms a locked condition. This process of locking the loop is called the capture effect.
- The time required for VCO to adjust its frequency with the signal frequency is called capture time. It depends on the internal parameters of the system. There is a limit for the input signal for which the system can acquire a locked condition. This range of frequency between which the system can goes into locked condition is called capture range. This range is symmetrical about centre frequency. This capture range depends upon filter and amplifier characteristics.
- If system acquires a locked condition then even if the signal frequency changes the loop remains in locked condition. The range of input frequency over which the locked condition maintained is called locked range. This also depends on amplifier and filter characteristics.

- The above operations take place until the VCO frequency equals the input signal frequency. Based on the type of application, we can use either the output of an active low-pass filter or the output of a VCO. PLLs are used in many **applications** such as FM demodulators, clock generator etc.
- PLL operates in one of the **following three modes** −
- Free running mode
- Capture mode
- Lock mode

- Initially, the PLL operates in **free running mode** when no input is applied to it. When an input signal having some frequency is applied to the PLL, then the output signal frequency of the VCO will start change.
- At this stage, the PLL is said to be operating in the **capture mode**. The output signal frequency of the VCO will change continuously until it is equal to the input signal frequency. Now, it is said that PLL is operating in the **lock mode**.

IC 565 Pin Diagram:

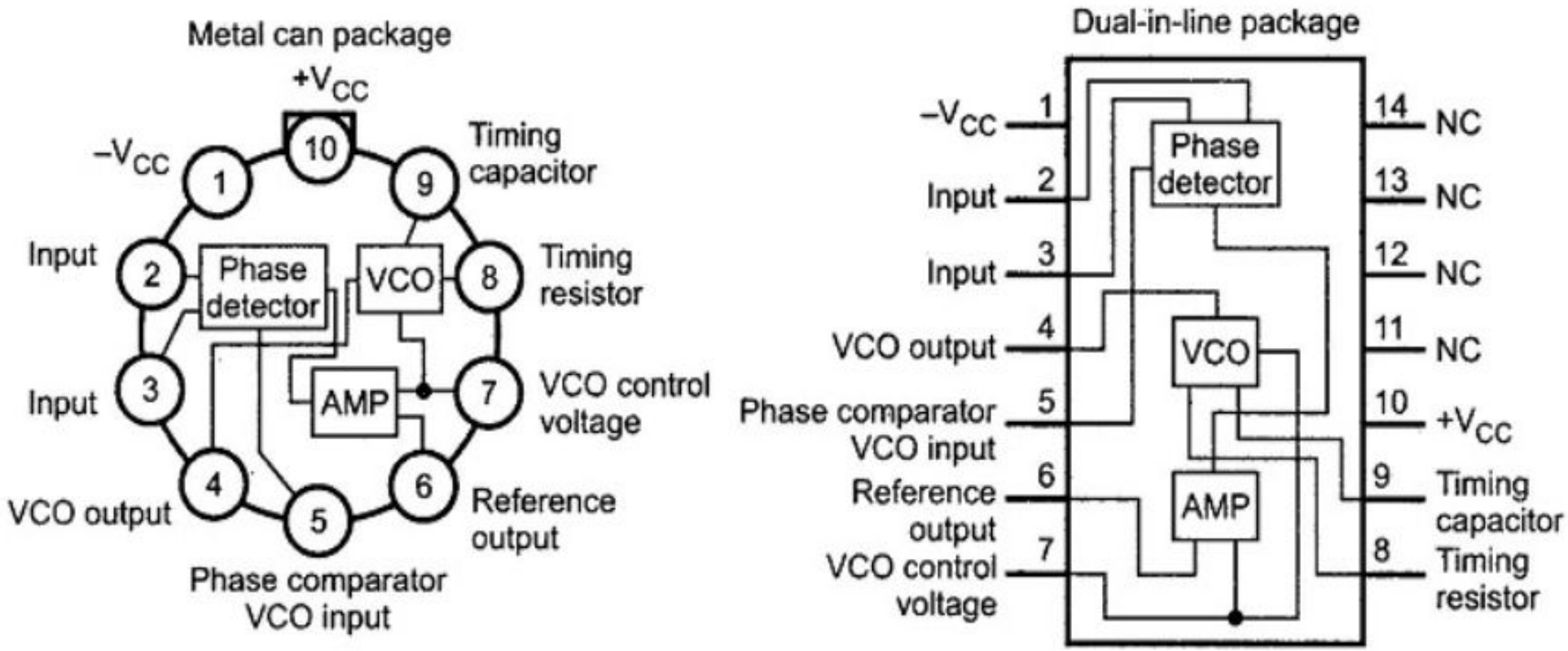


Fig. Pin configuration of IC 565

- **IC 565 PLL Block Diagram:**
- The block diagram of IC 565 PLL consists of a phase detector, amplifier, low pass filter, and VCO.
- Therefore, it is necessary to connect the output of the VCO (pin 4) to the phase comparator input (pin 5) externally.
- In frequency multiplication applications, a digital frequency divider is inserted into the loop, i.e., between pin 4 and pin 5.

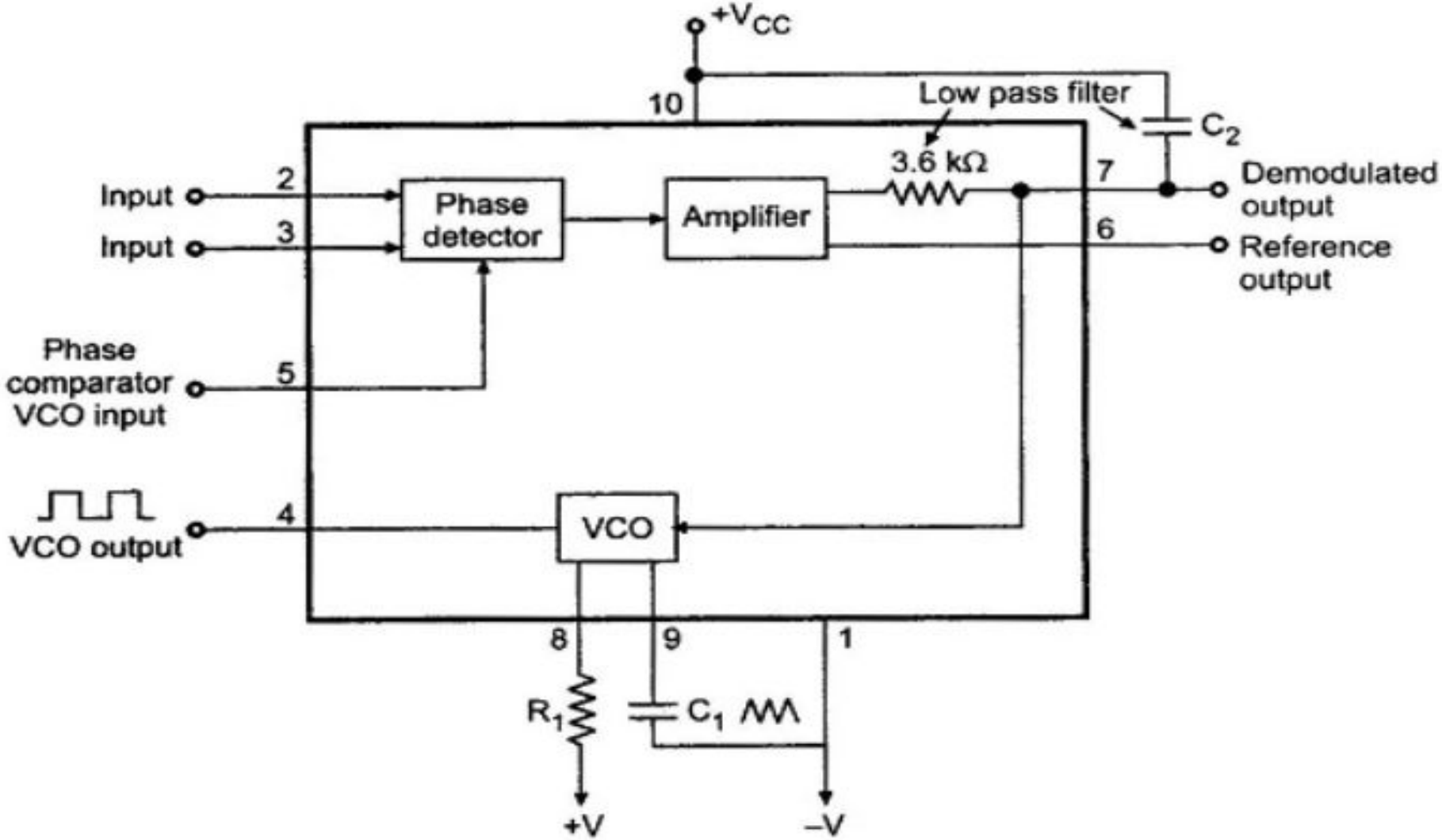


Fig. Block diagram of IC 565 PLL

- The centre frequency of the PLL is determined by the free-running frequency of the VCO, and it is given as,

$$f_0 = \frac{1.2}{4 R_1 C_1} \text{ Hz} \qquad \ldots (1)$$

where R1 and C1 are an external resistor and a capacitor connected to pins 8 and 9, respectively. The values of R1 and C1 are adjusted such that the free-running frequency will be at the centre of the input frequency range.

The value of R1 is restricted from 2kΩ to 20 kΩ but a capacitor can have any value. A capacitor C2 connected between pin 7 and the positive supply (pin 10) forms a first-order low-pass filter with an internal resistance of 3.6 kΩ.

The value of filter capacitor C2 should be large enough to eliminate possible oscillations in the VCO voltage.

- The lock range and capture range for IC 565 PLL are given by the following equations :

$$f_L = \pm \frac{8 f_o}{V} \text{ Hz} \qquad \ldots (2)$$

where

f_o = free running frequency of VCO in Hz

and

$V = (+V) - (-V)$ volts

$$f_c = \pm \left[\frac{f_L}{2\pi (3.6) (10^3) (C_2)} \right]^{1/2} \qquad \ldots (3)$$

where C_2 is in farads.

- where C_2 is in farads.
- From equation 2 we can notice that lock range increases with an increase in input voltage but decreases with increase in supply voltage.
- The two inputs (pin 2 and pin 3) to the phase detector allows direct coupling of an input signal, provided that there is no dc voltage difference between the pins and, the dc resistances seen from pins 2 and 3 are equal.
- A reference voltage at pin 6 is approximately equal to the dc voltage of the demodulated output at pin 7. This reference voltage may be used as a comparator input in applications like frequency shift keying.